University of Ghana
Department of Statistics and Actuarial Science
ACTU 301: Life Contingencies I

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EXERCISE 3, 4 and 5

QUE 1.

You are given:

(i) T_x and T_y are independent.

(ii) The survival function for (x) follows $l_x = 100(95 - x), 0 \leq x \leq 95$.

(iii) The survival function for (y) is based on a constant force of mortality, $\mu_{y+t} = \mu, t \geq 0$.

(iv) $n < 95 - x$

Determine the probability that (x) dies within n years and also dies before (y).

QUE 2.

You are given:

(i) Z is the present value random variable for an insurance on the lives of (x) and (y), where

$$Z = \begin{cases} T_y, & T_x \leq T_y \\ 0, & T_x > T_y \end{cases}$$

(ii) (x) is subject to a constant force of mortality, 0.07.

(iii) (y) is subject to a constant force of mortality, 0.09

(iv) (x) and (y) are independent lives.

(v) $\delta = 0.06$

Calculate E[Z].

QUE 3.

For a triple decrement table, you are given:

(i) Each decrement is uniformly distributed over each year of age in its associated single decrement table.

(ii) $q_x'^{(1)} = 0.200$

(iii) $q_x'^{(2)} = 0.080$

(iv) $q_x'^{(3)} = 0.125$

Calculate $q_x^{(1)}$.

QUE 4.

For a whole life insurance of 1000 on (80), with death benefits payable at the end of the year of death, you are given:

(i) Mortality follows a select and ultimate mortality table with a one-year select period.

(ii) $q_{[80]} = 0.5 q_{80}$

(iii) $i = 0.06$

(iv) $1000 A_{80} = 679.80$

(v) $1000 A_{81} = 689.52$

Calculate $1000 A_{[80]}$.

QUE 5.

A population of 1000 lives age 60 is subject to 3 decrements, death (1), disability (2), and retirement (3). You are given:

(i) The following independent rates of decrement:

x	$q_x'^{(1)}$	$q_x'^{(2)}$	$q_x'^{(3)}$
60	0.010	0.030	0.100
61	0.013	0.050	0.200

(ii) Decrements are uniformly distributed over each year of age in the multiple decrement table.

Calculate the expected number of people who will retire before age 62.

QUE 6.

In a population, non-smokers have a force of mortality equal to one half that of smokers.

For non-smokers, $l_x = 500(110 - x), \quad 0 \le x \le 110$.

Calculate $\mathring{e}_{20:25}$ for a smoker (20) and a non-smoker (25) with independent future lifetimes.

QUE 7.

For a whole life insurance of 1 on (x) with benefits payable at the moment of death, you given:

(i) δ_t, the force of interest at time t is $\delta_t = \begin{cases} 0.02, & t < 12 \\ 0.03, & t \ge 12 \end{cases}$

(ii) $\mu_{x+t} = \begin{cases} 0.04, & t < 5 \\ 0.05, & t \ge 5 \end{cases}$

Calculate the actuarial present value of this insurance.

QUE 8.

You are given the survival function

$$S_0(t) = 1 - (0.01t)^2, \qquad 0 \le t \le 100$$

Calculate $\mathring{e}_{30:\overline{50}|}$, the 50-year temporary complete expectation of life of (30).

QUE 9.

For a special 3-year term insurance on (x), you are given:

(iv) Z is the present-value random variable for this insurance.

(v) $q_{x+k} = 0.02(k+1), \qquad k = 0, 1, 2$

(vi) The following benefits are payable at the end of the year of death:

k	b_{k+1}
0	300
1	350
2	400

(iv) $i = 0.06$

Calculate $\text{Var}(Z)$.

QUE 10.

For a double decrement table, you are given:

Age	$l_x^{(\tau)}$	$d_x^{(1)}$	$d_x^{(2)}$
40	1000	60	55
41	–	–	70
42	750	–	–

Each decrement is uniformly distributed over each year of age in the double decrement table.

Calculate $q_{41}'^{(1)}$.

QUE 11.

, For a special whole life insurance on (x), you are given:

(i) Z is the present value random variable for this insurance.

(ii) Death benefits are paid at the moment of death.

(iii) $\mu_{x+t} = 0.02, \quad t \geq 0$

(iv) $\delta = 0.08$

(v) The death benefit at time t is $b_t = e^{0.03t}, \quad t \geq 0$

Calculate $\mathrm{Var}(Z)$.

QUE 12.

You are given:

$$\mu_x = \begin{cases} 0.05 & 50 \leq x < 60 \\ 0.04 & 60 \leq x < 70 \end{cases}$$

Calculate ${}_{4|14}q_{50}$.

QUE 13.

For a life table with a one-year select period, you are given:

(i)

x	$l_{[x]}$	$d_{[x]}$	l_{x+1}	$\mathring{e}_{[x]}$
80	1000	90	–	8.5
81	920	90	–	–

(ii) Deaths are uniformly distributed over each year of age.

Calculate $\mathring{e}_{[81]}$.

QUE 14.

You are given:

(i) $\overset{\circ}{e}_{30:\overline{40}|} = 27.692$

(ii) $S_0(t) = 1 - \dfrac{t}{\omega}, \qquad 0 \le t \le \omega$

(iii) T_x is the future lifetime random variable for (x).

Calculate $\operatorname{Var}(T_{30})$.

QUE 15.

A fully discrete 3-year term insurance of 10,000 on (40) is based on a double-decrement model, death and withdrawal:

(i) Decrement 1 is death.

(ii) $\mu_{40+t}^{(1)} = 0.02, \quad t \ge 0$

(iii) Decrement 2 is withdrawal, which occurs at the end of the year.

(iv) $q'^{(2)}_{40+k} = 0.04, \qquad k = 0, 1, 2$

(v) $v = 0.95$

Calculate the actuarial present value of the death benefits for this insurance.

QUE 16.

For independent lives (50) and (60):

$$\mu_x = \frac{1}{100 - x}, \qquad 0 \le x < 100$$

Calculate $\overset{\circ}{e}_{\overline{50:60}}$.

QUE 17.

For a double decrement table with $l_{40}^{(\tau)} = 2000$:

x	$q_x^{(1)}$	$q_x^{(2)}$	$q_x'^{(1)}$	$q_x'^{(2)}$
40	0.24	0.10	0.25	y
41	--	--	0.20	$2y$

Calculate $l_{42}^{(\tau)}$.

QUE 18.

You are given the following extract from a select-and-ultimate mortality table with a 2-year select period:

x	$l_{[x]}$	$l_{[x]+1}$	l_{x+2}	$x+2$
60	80,625	79,954	78,839	62
61	79,137	78,402	77,252	63
62	77,575	76,770	75,578	64

Assume that deaths are uniformly distributed between integral ages.

Calculate ${}_{0.9}q_{[60]+0.6}$.

QUE 19.

For a multiple decrement table, you are given:

(i) Decrement (1) is death, decrement (2) is disability, and decrement (3) is withdrawal.

(ii) $q_{60}'^{(1)} = 0.010$

(iii) $q_{60}'^{(2)} = 0.050$

(iv) $q_{60}'^{(3)} = 0.100$

(v) Withdrawals occur only at the end of the year.

(vi) Mortality and disability are uniformly distributed over each year of age in the associated single decrement tables.

Calculate $q_{60}^{(3)}$.

QUE 20.

For independent lives (x) and (y):

(i) $q_x = 0.05$

(ii) $q_y = 0.10$

(iii) Deaths are uniformly distributed over each year of age.

Calculate ${}_{0.75}q_{xy}$.

QUE 21.

For a double-decrement model:

(i) ${}_tp'^{(1)}_{40} = 1 - \frac{t}{60}, \qquad 0 \le t \le 60$

(ii) ${}_tp'^{(2)}_{40} = 1 - \frac{t}{40}, \qquad 0 \le t \le 40$

Calculate $\mu^{(\tau)}_{40+20}$.

QUE 22.

For a population of individuals, you are given:

(i) Each individual has a constant force of mortality.

(ii) The forces of mortality are uniformly distributed over the interval (0,2).

Calculate the probability that an individual drawn at random from this population dies within one year.

QUE 23.

For a special 3-year term insurance on (x), you are given:

(i) Z is the present-value random variable for the death benefits.

(ii) $q_{x+k} = 0.02(k+1) \qquad k = 0, 1, 2$

(iii) The following death benefits, payable at the end of the year of death:

k	b_{k+1}
0	300,000
1	350,000
2	400,000

(iv) $i = 0.06$

Calculate $\mathrm{E}(Z)$.

QUE 24.

Z is the present value random variable for a 15-year pure endowment of 1 on (x):

(i) The force of mortality is constant over the 15-year period.

(ii) $v = 0.9$

(iii) $\mathrm{Var}(Z) = 0.065\,\mathrm{E}[Z]$

Calculate q_x.

QUE 25.

For a double decrement model:

(i) In the single decrement table associated with cause (1), $q'^{(1)}_{40} = 0.100$ and decrements are uniformly distributed over the year.

(ii) In the single decrement table associated with cause (2), $q'^{(2)}_{40} = 0.125$ and all decrements occur at time 0.7.

Calculate $q^{(2)}_{40}$.

QUE 26.

Don, age 50, is an actuarial science professor. His career is subject to two decrements:

(i) Decrement 1 is mortality. The associated single decrement table follows $l_x = 100 - x, 0 \le x \le 100$.

(ii) Decrement 2 is leaving academic employment, with

$$\mu_{50+t}^{(2)} = 0.05, \quad t \ge 0$$

Calculate the probability that Don remains an actuarial science professor for at least five but less than ten years.

QUE 27.

You are given:

$$\mu_x = \begin{cases} 0.04, & 0 < x < 40 \\ 0.05, & x \ge 40 \end{cases}$$

Calculate $\mathring{e}_{25:\overline{25}|}$.

QUE 28.

For a whole life insurance of 1 on (x), you are given:

(i) The force of mortality is μ_{x+t}.

(ii) The benefits are payable at the moment of death.

(iii) $\delta = 0.06$

(iv) $\bar{A}_x = 0.60$

Calculate the revised expected present value of this insurance assuming μ_{x+t} is increased by 0.03 for all t and δ is decreased by 0.03.

QUE 29.

For a double-decrement table where cause 1 is death and cause 2 is withdrawal, you are given:

(i) Deaths are uniformly distributed over each year of age in the single-decrement table.

(ii) Withdrawals occur only at the end of each year of age.

(iii) $l_x^{(\tau)} = 1000$

(iv) $q_x^{(2)} = 0.40$

(v) $d_x^{(1)} = 0.45\ d_x^{(2)}$

Calculate $p_x'^{(2)}$.

QUE 30.

For a double decrement table, you are given:

(i) $q'^{(1)}_x = 0.2$

(ii) $q'^{(2)}_x = 0.3$

(iii) Each decrement is uniformly distributed over each year of age in the double decrement table.

Calculate ${}_{0.3}q_{x+0.1}^{(1)}$.

QUE 31.

You are given:

(i) the following select-and-ultimate mortality table with 3-year select period:

x	$q_{[x]}$	$q_{[x]+1}$	$q_{[x]+2}$	q_{x+3}	$x+3$
60	0.09	0.11	0.13	0.15	63
61	0.10	0.12	0.14	0.16	64
62	0.11	0.13	0.15	0.17	65
63	0.12	0.14	0.16	0.18	66
64	0.13	0.15	0.17	0.19	67

(ii) $i = 0.03$

Calculate ${}_{2|2}A_{[60]}$, the actuarial present value of a 2-year deferred 2-year term insurance on $[60]$.

QUE 32.

For a triple decrement table, you are given:

(i) $\mu_{x+t}^{(1)} = 0.3,\ t > 0$

(ii) $\mu_{x+t}^{(2)} = 0.5,\ t > 0$

(iii) $\mu_{x+t}^{(3)} = 0.7,\ t > 0$

Calculate $q_x^{(2)}$.

QUE 33.

You are given:

(i) $l_x = 10(105 - x),\ 0 \le x \le 105$.

(ii) (45) and (65) have independent future lifetimes.

Calculate $\mathring{e}_{\overline{45:65}}$.

QUE 34.

Given: The survival function $S_0(t)$, where

$$S_0(t) = 1, \qquad 0 \le t < 1$$

$$S_0(t) = 1 - \left\{ \left(e^x\right)/100 \right\}, \qquad 1 \le t < 4.5$$

$$S_0(t) = 0, \qquad 4.5 \le x$$

Calculate μ_4.

QUE 35.

For a double decrement table, you are given:

(i) $\mu_{x+t}^{(1)} = 0.2\, \mu_{x+t}^{(\tau)}, \quad t > 0$ and instruction requested lower case tau, but it already was

(ii) $\mu_{x+t}^{(\tau)} = kt^2, \quad t > 0$

(iii) $q_x^{\prime(1)} = 0.04$

Calculate ${}_2q_x^{(2)}$.

QUE 36.

For a whole life insurance of 1 on (41) with death benefit payable at the end of year of death, you are given:

(i) $i = 0.05$

(ii) $p_{40} = 0.9972$

(iii) $A_{41} - A_{40} = 0.00822$

(iv) ${}^2A_{41} - {}^2A_{40} = 0.00433$

(v) Z is the present-value random variable for this insurance.

Calculate Var(Z).

QUE 37.

For a special whole life insurance on (x), payable at the moment of death:

(i) $\mu_{x+t} = 0.05\,,\ \ t > 0$

(ii) $\delta = 0.08$

(iii) The death benefit at time t is $b_t = e^{0.06t}\,,\ \ t > 0$.

(iv) Z is the present value random variable for this insurance at issue.

Calculate $\mathrm{Var}(Z)$.

QUE 38.

For two independent lives now age 30 and 34, you are given:

x	q_x
30	0.1
31	0.2
32	0.3
33	0.4
34	0.5
35	0.6
36	0.7
37	0.8

Calculate the probability that the last death of these two lives will occur during the 3rd year from now (i.e. ${}_{2|}q_{\overline{30:34}}$).

QUE 39.

For a special whole life insurance on (x), you are given:

(i) Z is the present value random variable for this insurance.

(ii) Death benefits are paid at the moment of death.

(iii) $\mu_{x+t} = 0.02, \quad t \geq 0$

(iv) $\delta = 0.08$

(v) The death benefit at time t is $b_t = e^{0.03t}, \quad t \geq 0$

Calculate $\text{Var}(Z)$.

QUE 40.

For a special whole life insurance policy issued on (40), you are given:

(i) Death benefits are payable at the end of the year of death.

(ii) The amount of benefit is 2 if death occurs within the first 20 years and is 1 thereafter.

(iii) Z is the present value random variable for the payments under this insurance.

(iv) $i = 0.03$

(v)

x	A_x	${}_{20}E_x$
40	0.36987	0.51276
60	0.62567	0.17878

(vi) $E\left[Z^2\right] = 0.24954$

Calculate the standard deviation of Z.

QUE 41.

You are given:

(i) $q_{60} = 0.01$

(ii) Using $i = 0.05$, $A_{60:\overline{3}|} = 0.86545$.

Using $i = 0.045$, calculate $A_{60:\overline{3}|}$.

QUE 42.

You are given:

(i) $A_{35:\overline{15}|} = 0.39$

(ii) $A^{\ 1}_{35:\overline{15}|} = 0.25$

(iii) $A_{35} = 0.32$

Calculate A_{50}.

QUE 43.

You are given:

(i) Z_1 is the present value random variable for an *n*-year term insurance of 1000 issued to (*x*).

(ii) Z_2 is the present value random variable for an *n*-year endowment insurance of 1000 issued to (*x*).

(iii) For both Z_1 and Z_2 the death benefit is payable at the end of the year of death.

(iv) $E[Z_1] = 528$

(v) $Var(Z_2) = 15{,}000$

(vi) $A_{x:\overset{1}{\overline{n}|}} = 0.209$

(vii) ${}^2A_{x:\overset{1}{\overline{n}|}} = 0.136$

Calculate $Var(Z_1)$.

QUE 44.

For three fully discrete insurance products on the same (*x*), you are given:

(i) Z_1 is the present value random variable for a 20-year term insurance of 50.

(ii) Z_2 is the present value random variable for a 20-year deferred whole life insurance of 100.

(iii) Z_3 is the present value random variable for a whole life insurance of 100.

(iv) $E[Z_1] = 1.65$ and $E[Z_2] = 10.75$.

(v) $Var(Z_1) = 46.75$ and $Var(Z_2) = 50.78$.

Calculate $Var(Z_3)$.

QUE 45.

For a 2-year deferred, 2-year term insurance of 2000 on [65], you are given:

(i) The following select and ultimate mortality table with a 3-year select period:

x	$q_{[x]}$	$q_{[x]+1}$	$q_{[x]+2}$	q_{x+3}	$x+3$
65	0.08	0.10	0.12	0.14	68
66	0.09	0.11	0.13	0.15	69
67	0.10	0.12	0.14	0.16	70
68	0.11	0.13	0.15	0.17	71
69	0.12	0.14	0.16	0.18	72

(ii) $i = 0.04$

(iii) The death benefit is payable at the end of the year of death.

Calculate the actuarial present value of this insurance.

QUE 46.

For a special whole life insurance on (x), you are given:

(i) Death benefits are payable at the moment of death.

(ii) The death benefit at time t is $b_t = e^{0.02t}$, for $t \geq 0$.

(iii) $\mu_{x+t} = 0.04$, for $t \geq 0$

(iv) $\delta = 0.06$

(v) Z is the present value at issue random variable for this insurance.

Calculate $Var(Z)$.

QUE 47.

You are given the following extract of ultimate mortality rates from a two-year select and ultimate mortality table:

x	q_x
50	0.045
51	0.050
52	0.055
53	0.060

The select mortality rates satisfy the following:

(i) $q_{[x]} = 0.7q_x$

(ii) $q_{[x]+1} = 0.8q_{x+1}$

You are also given that $i = 0.04$.

Calculate $A^{1}_{[50]:\overline{3}|}$.

QUE 48.

For a 25-year pure endowment of 1 on (x), you are given:

(i) Z is the present value random variable at issue of the benefit payment.

(ii) $Var(Z) = 0.10E[Z]$

(iii) ${}_{25}p_x = 0.57$

Calculate the annual effective interest rate.

QUE 49.

For a double decrement table, you are given:

(i) $q'^{(1)}_x = 0.1$

(ii) $q^{(2)}_x = 0.2$

(iii) Each decrement is uniformly distributed over each year of age in its associated single decrement table.

Calculate $q^{(1)}_x$.

QUE 50.

You are given the following excerpt from a double decrement table:

x	$\ell^{(\tau)}_x$	$q^{(1)}_x$	$q^{(2)}_x$
53	---	0.025	0.030
54	5000	---	0.040
55	4625	0.055	0.050

Calculate ${}_2q^{(1)}_{53}$.

QUE 51.

You are given the following extract from a triple decrement table:

x	$\ell^{(\tau)}_x$	$q^{(1)}_x$	$q^{(2)}_x$	$q^{(3)}_x$
40	15,000	0.01	0.04	0.05
41	-	0.04	0.08	0.10

After the table was prepared, you discover that $q^{(1)}_{40}$ should have been 0.02, and that all other numerical values shown above are correct.

Calculate the resultant change in $d^{(3)}_{41}$.

QUE 52.

For two lives, (80) and (90), with independent future lifetimes, you are given:

k	p_{80+k}	p_{90+k}
0	0.9	0.6
1	0.8	0.5
2	0.7	0.4

Calculate the probability that the last survivor will die in the third year.

QUE 53.

For two lives, (60) and (70), with independent future lifetimes, you are given:

(i) $_5p_{60} = 0.92$

(ii) $_5p_{70} = 0.88$

(iii) $1000\, q_{65} = 21.32$

(iv) $1000\, q_{75} = 51.69$

Calculate $1000\ {}_{5|}q_{\overline{60:70}}$.